ALKANE

1. (d) C_7H_{16} (C_nH_{2n+2})

2. (a) n-Pentane > iso-Pentane > neo-Pentane

   Explanation:

   Boiling point of alkanes depends on **surface area** and **van der Waals (dispersion) forces**.

   - More surface area → Stronger intermolecular forces → Higher boiling point
   - More branching → Less surface area → Lower boiling point

3. (a) According to wurtz reaction.

$$2CH_3CH_2CH_2Br + 2Na \xrightarrow{\text{ether}} CH_3(CH_2)_4CH_3 + 2NaBr$$

4. (a) Mg

   Explanation:

   A **Grignard reagent** is prepared by reacting a **haloalkane** (R – X) with **magnesium metal (Mg)** in the presence of **dry ether**.

5. (b) $\underset{\text{Sodium acetate}}{2CH_3COONa} + 2H_2O \xrightarrow{\text{Electolysis}} CH_3 - CH_3 + 2CO_2 + 2NaOH + H_2$

6. (c) $Pb(C_2H_5)_4 \xrightarrow{\text{heat}} Pb + \underset{\text{Ethyl radical}}{4CH_3CH_2}$

$$\underset{Br}{\underset{|}{CH_2}} - \underset{Br}{\underset{|}{CH_2}} + Pb \longrightarrow \underset{\text{Ethene}}{CH_2 = CH_2} + \underset{\text{Lead bromide}}{PbBr_2}$$

   As leaded gasoline burns, lead metal gets deposited in the engine which is removed by adding ethylene dibromide. The lead bromide is volatile and is carried off with the exhaust gases from the engine

7. (a) $C_{15} - C_{18}$

   Explanation:

   Petroleum is separated into fractions based on **boiling point range** and **number of carbon atoms**.

Fraction	Carbon Range
Petroleum gas	$C_1 - C_4$
Petrol (Gasoline)	$C_5 - C_9$
Naphtha	$C_6 - C_{10}$
Kerosene oil	**$C_{15} - C_{18}$**
Diesel	$C_{15} - C_{20}$
Lubricating oil	$C_{20}+$

Therefore, **kerosene oil = $C_{15} - C_{18}$**

8. **(a) Wurtz reaction**

Explanation:

This reaction involves:

- Two alkyl halide molecules
- Sodium metal (Na)
- Dry ether

This is called the **Wurtz reaction**.

9. (d) $C_2H_5I + 2Na + IC_2H_5 \xrightarrow[\text{Ether}]{\text{Dry}} \underset{\text{Butane}}{C_2H_5 - C_2H_5} + 2NaI$

10. (c) $(CH_3)_3CH \xrightarrow{KMnO_4} \underset{\text{tertiary butyl alcohol}}{(CH_3)_3C - OH}$

11. (c) 3^0

12. **(c) Isobutane**

Concept:

Volatility $\propto$ 1 / Boiling Point

- Lower boiling point → More volatile
- More branching → Lower boiling point
- Fewer carbon atoms → Lower boiling point

13. (c) $RCl + 2Na + RCl \xrightarrow[Ether]{Dry} 2NaCl + \underset{\text{Alkane}}{R-R}$

14. (d) *iso*-octane *i.e.* 2,2,4-trimethyl pentane has highest octane number.

15. (b) Refrigerant

What is Freon-12?

Freon-12 is the trade name for Dichlorodifluoromethane.

- Chemical formula: CCl_2F_2
- Also called R-12
- Belongs to chlorofluorocarbons (CFCs)

16. (a) 20% normal heptane + 80% iso-octane

Concept of Octane Number

The **octane number** measures the **anti-knocking property** of petrol.

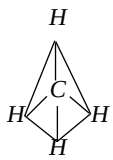
- n-Heptane → Octane number = 0 (poor anti-knocking)
- Iso-octane → Octane number = 100 (excellent anti-knocking)

Octane number of petrol = % of iso-octane in a mixture of iso-octane and n-heptane having the same knocking property.

17. (b) With calculated amount of Grignard reagent, acetyl chloride forms ketones.

$$CH_3COCl + XMgCH_3 \rightarrow CH_3COCH_3 + Mg\begin{matrix} Cl \\ X \end{matrix}$$

18. (d) CH_4 is tetrahedral



sp^3 hybridization (tetrahedra l)

19. (b) 2,2-dimethylpropane

Concept:

A compound will give only one monobrominated product if all hydrogen atoms are equivalent.

20. (a) Less volatile

Explanation:

Kerosene has:

- Moderate boiling point
- Less volatility compared to petrol
- Burns smoothly without rapid evaporation

If it were highly volatile, it would evaporate quickly and be unsafe.

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